

Asmaa Abdul-Amin

asmaaabdulamin@gmail.com | 240-308-2269 | github.com/blkpvnthr | Clarksburg, MD

Summary

Software Engineer with 2+ years of experience building production software, REST APIs, backend systems, and automation tools supporting NASA mission platforms at The Johns Hopkins University Applied Physics Laboratory. Experienced across the full software development lifecycle using Python, PHP, JavaScript, SQL, and cloud-native development practices. **Previously held an Interim Secret clearance** (currently inactive and eligible for reactivation through a sponsoring employer).

Skills

- Programming Languages: Python, PHP, JavaScript, React, SQL, JSON, HTML, XML, ReqIF, CSS
 - Frameworks & Technologies: Atlassian, Bootstrap, Elasticsearch, Filebeat, IBM DOORS, REST APIs, Linux, Git
 - Libraries & Tools: Pandas, NumPy, SciPy, Torch, lxml, gymnasium, INotify, TensorFlow, Hugging Face, OpenAI API, Qiskit
-

Experience

Johns Hopkins Applied Physics Laboratory

Software Engineering Intern

Space Exploration Sector (SOF-2) | May 2024 – May 2026

- Partnered with the project's principal investigator through recurring architecture sessions to engineer a Linux-compatible Python ETL pipeline that normalized IBM DOORS ReqIF/XML data, generated structured JSON, and indexed requirements and verification events in Elasticsearch for identifier-, keyword-, and relationship-based retrieval.
- Developed and maintained production web applications and SQL-backed administrative portals across 15+ NASA/APL mission platforms using PHP, JavaScript, SQL, Bootstrap, and DataTables.
- Built Python and SQL database-export utilities that generated JSON feeds for scientific analysis and integration with internal engineering systems.
- Modernized legacy NASA mission applications by migrating archived scientific data into APL databases, retiring obsolete sites, and rebuilding interfaces as maintainable APL-hosted applications.
- Owned delivery of the new JHUAPL Ice Giants website, migrating data from retired SQL databases and building automated countdowns, reusable event templates, LISTSERV subscriber workflows, and administrative tooling across the Ice Giants and various other APL platforms.
- Collaborated directly with scientists, stakeholders, and mission leadership to translate requirements into deployed solutions—including compliance-sensitive internal portals—and owned features through development, testing, deployment, and maintenance.

Undergraduate Research Intern

Research & Exploratory Development Department (R1N) | Mar 2023 – May 2024

- Deployed multiple webpages for the NASA Dragonfly internal portal, streamlining access to critical mission resources and improving mission collaboration efficiency.
 - Developed machine learning models analyzing COVID-19 policies vs. mortality, improving interpretability of public health data.
 - Presented research at NIST-PREP (Professional Research Experience Program) Showcase.
-

Projects

Parker Solar Probe - Developed and maintained the Parker-Kit Celebration webpage.

Ice Giants - Planned, estimated, and implemented the Ice Giants research website.

EZIE - Rebuilt the EZIE website from ReactJS to PHP.

LOGIC - Implemented a database-backed terminology management system for standardized lunar infrastructure definitions.

LSIC - Created JSON APIs and optimized database-backed newsletter management.

BLKPVNTHR.OS – Engineering an AI-powered personal operating system featuring specialized autonomous agents and workflows.

Education

B.S. Data Science – University of Maryland Global Campus (Expected Dec 2028)

A.A. Computer Science – Montgomery College (Expected Dec 2026)

Certifications:

IBM – Cyber Defense Architecture (2026)

The Johns Hopkins University Applied Physics Laboratory – Python Programming II (2025)

Univ. of Chicago – Quantum Computing Systems Design I (2024)

IBM – Machine Learning with Python (2023)

Harvard – CS50 Intro to Computer Science (2022)